

Date: 9th March 26

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F2023-744

Sun M/Tue Wed Thu Fri Sat

ToA (Assignment 1) Sec C

Question 1.

0 appears tripled (3, 6, 9, ...)

Sol: aaa, aaaa

RE $(aaa)^*$

b can appear anywhere $(b + aaa)^*$

Question 2

At least one of the strings s_1, s_2, s_3 or s_4

Sol: $\Sigma^* (s_1 + s_2 + s_3 + s_4) \Sigma^*$

Question 3

(a) All words with exactly 3 bs

Sol: $a^* b a^* b a^* b a^*$

(b) strings with exactly 2 or 3 bs

Sol: $a^* b a^* b a^* + a^* b a^* b a^* b a^*$

Question 4

(a) strings that end in double letter

(-) aa (-) bb

Sol: RE = $(a + b)^* (aa + bb)$

(b) strings with one double letter

RE = $(a + b)^* (aa + bb) (a + b)^*$

Question 5



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a, ab, bb, bab, bbb

String where b never tripples

Sol: RE $(a+ba+bb^2)^*$

Question 6

a is tripled or b is tripled, not both

Sol: $aaa \times bbb$
 $bbb \times aaa$

$[(a+b)^* aaa (a+b)^*] + [(a+b)^* bbb (a+b)^*]$

Question 7

$\Sigma = \{a, b\}$

(a) substring ab

Sol: $a^* b^*$

(b) do not have bba and abb

Sol: $(a+b)^* - (a+b)^* bba (a+b)^* - bba (a+b)^*$
 $(a+b)^* - (a+b)^* abb (a+b)^* - abb (a+b)^*$

RE $(a+b)^* - (a+b)^* bba (a+b)^* + ((a+b)^* - (a+b)^* abb (a+b)^*)$

Question 8

Total no of a's is divisible by 3 such as aabbaabbaa

Sol: aaa

$(b^* ab^* ab^* ab^*)^*$

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Question 9

(a) b's occur in odd clamps

sol: bb^*

$b(bb)^*$ odd

$\Rightarrow (a + b(bb)^*)^*$

(b) even no of a's odd no of b's

sol: $(aa)^* b(bb)^* \rightarrow \text{odd}$

$\Rightarrow (aa)^* b(bb)^*$

(c) odd no of a's and odd no of b's

sol: $a(aa)^* \text{ odd } (b(bb)^*)^*$

$\Rightarrow a(aa)^* b(bb)^*$

Question 10

$(a+b)^* a(a+b)^* b(a+b)^*$

(a) $(a+b)^* ab(a+b)^*$

sol: $(a+b)^* a(a+b)^* b(a+b)^*$

lang of RE 1 $ab, aab, abb, abab, baab, aaabb, abba$

RE 2 $ab, aab, abb, abab, baab, aaabb$

so

$(a+b)^* a(a+b)^* b(a+b)^* = (a+b)^* ab(a+b)^*$

(b) $(a+b)^* ab(a+b)^* + b^* a^* = (a+b)^*$

sol: RE 1: $(a+b)^* ab(a+b)^*$

lang: $ab, aab, abb, abab, baab$

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part 2: $b^* a^*$

lang: $\Lambda, b, bb, a, ba, bba, bbaa$

$$L = (a+b)^*$$

$$RE_2 (a+b)^*$$

$$L = (a+b)^*$$

$$(a+b)^* = (a+b)^*$$

$$(c) (a+b)^* ab [(a+b)^* ab (a+b)^* + b^* a^*] + b^* a^* = (a+b)^*$$

Sol: Already proved in part b

$$(a+b)^*$$

all possible string

Question 11

(a) $(ab)^* a$ and $a(ba)^*$

Sol: string	$(ab)^* a$	$a(ba)^*$	
a	✓	✓	} both generate a, abq, ababaq, ...
aba	✓	✓	
ababaq	✓	✓	

equivalent

(b) $(a^* + b)^*$ and $(a+b)^*$

Sol: a^* — only a

b generates b

produce all strings equivalent

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(c) $(a^* + b^*)^*$ and $(a+b)^*$

Sol: $a^*, b^*, aa, bbb, abab \Rightarrow$ equivalent

(d) ϵ^* and ϵ

Sol: $\epsilon^* = \{\epsilon\} \Rightarrow$ equivalent

(e) $(ab)a$ vs $a(ba^*)^*$

Sol: both generate any no of a separated by $b \Rightarrow$ equivalent

(f) $(a^* bbb)^* a^*$ and $a^* (bbba^*)^*$

Sol: groups of bbb separated by $a \Rightarrow$ equivalent

(g) $(a+bb)aa$ vs $\epsilon + (a+bb)^* aa$

Sol: first expression: zero or more repetitions

second expression: either ϵ or single repetition

$\Rightarrow$ equivalent

(h) $(aa)^* (\epsilon + a)$ and a^*

Sol: $(aa)^*$ even $(\epsilon + a) \Rightarrow$ equivalent

(i) $a(ab)^* (\epsilon + a) b + b$ and $a^* b$

Sol: first: $b, ab, aab, abab, abaab, \dots$

second: b, ab, aab, aab, aab [NOT equivalent]

(j) $a(ba + a^* a)b$ and $aa^* b(aa^* b)^*$

Sol: first: $ab, aab, aaab, aaab, abab, aabab$

second: $ab, aab, aaab, aaab, abab, aabab$ [equivalent]

(k) $1 + a(a+b)^* + (a+b)^* aa(a+b)^*$ and $(1b^*) aab^*$

Sol: first: $1, a, ab, aba, abb, aa, baa, bbaa, babaa$

second: $1, a, ab, ba, abba, baab, aa, baa, baba$

[NOT equivalent]

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Question 12

(a) $(a+b)^* a (c+bbbb)$

Sol: All strings over alphabet that end with acbbb

lang: acbbb, aacbbb, bacbbb

(b) $(aaa)^* (bbb)^*$

Sol: all strings having any no of aaa block followed by any no of bbb

lang: Λ , aaa, bbb, aaabbb, aaaaabbbbbb, ...

(c) $b(bbb)^* a(aaa)^* b(bb)^* (aaa)^*$

Sol: Start with b, have groups of bbb, one a, groups of aa, one b, bb aaa lang: bab, babaa, bbbbabbb, ...

(d) $b(bbb)^* a(aa)^*$

Sol: Start with 1 b, any no of bbb, 1 a any no of aa

lang: ba, bba, bbaaa, bbbba, ...

(e) $b(bbb)^* a(aa)^* b(bb)^* (aaa)^*$

Done above

(f) $((a+b)a)^*$

Sol: all strings made by repeating 2-letter block

whose 2nd letter is always a

lang: Λ , aa, ba, aaba, baaa, babaa, ...

(b) $a(a+bbb)^*$

Sol: all string made by repeating a or bbb any no of times

lang: Λ , a, bbb, aa, abbb, bbbba, ...